

Table 28. Current consumption in Run mode from flash memory at different die temperatures

Symbol	Parameter	Conditions			Typ				Max ⁽¹⁾				Unit
		General ⁽²⁾	f _{HCLK}	Fetch from ⁽³⁾	25 °C	85 °C	105 °C	125 °C	25 °C	85 °C	105 °C	125 °C	
I _{DD(Run)}	Supply current in Run mode	f _{HCLK} = f _{HSE_bypass} (> 32.768 kHz), f _{HCLK} = f _{LSE_bypass} (= 32.768 kHz)	48 MHz	Flash memory	3.05	3.15	3.25	3.35	3.60	3.80	4.10	4.60	mA
			32 MHz		2.10	2.15	2.25	2.35	2.50	2.70	3.00	3.50	
			24 MHz		1.80	1.85	1.90	2.05	2.10	2.40	2.70	3.20	
			16 MHz		1.25	1.30	1.35	1.45	1.50	1.70	2.00	2.50	
			8 MHz		0.655	0.710	0.765	0.865	0.790	1.10	1.40	1.90	
			4 MHz		0.3654	0.420	0.470	0.570	0.460	0.700	0.980	1.50	
			2 MHz		0.225	0.270	0.325	0.425	0.290	0.540	0.820	1.40	
			1 MHz		0.150	0.200	0.250	0.350	0.200	0.450	0.730	1.30	
			500 kHz		0.115	0.160	0.215	0.315	0.160	0.410	0.690	1.20	
			125 kHz		0.0875	0.135	0.185	0.285	0.130	0.380	0.650	1.20	
			32.768 kHz		0.082	0.130	0.180	0.280	0.120	0.370	0.650	1.20	
		f _{HCLK} = f _{HSI48/HSIDIV} (> 32 kHz), f _{HCLK} = f _{LSI} (= 32 kHz)	48 MHz		3.40	3.50	3.55	3.60	3.90	4.10	4.40	4.90	
			24 MHz		2.25	2.30	2.35	2.45	2.60	2.80	3.10	3.60	
			12 MHz		1.45	1.50	1.55	1.65	1.70	1.90	2.20	2.70	
			6 MHz		1.05	1.10	1.15	1.20	1.20	1.40	1.70	2.20	
			3 MHz		0.855	0.880	0.925	1.00	0.960	1.20	1.50	2.00	
			1.5 MHz		0.750	0.780	0.825	0.915	0.840	1.10	1.40	1.90	
			750 kHz		0.700	0.730	0.775	0.865	0.780	1.00	1.30	1.80	
			375 kHz		0.675	0.705	0.750	0.840	0.760	0.970	1.30	1.80	
			32 kHz		0.082	0.130	0.180	0.280	0.120	0.370	0.650	1.20	

1. Evaluated by characterization – Not tested in production.
2. V_{DD} = 3.0 V for values in Typ columns and 3.6 V for values in Max columns, all peripherals disabled.
3. Prefetch disabled and cache enabled when fetching from flash memory.